

MSc

Carlos Canoyra Calero

Section

Energy Markets and Analytics

Division

Power and Energy Systems

Today's battery bids can reproduce a state-of-charge market

Equivalent interval bids for energy storage, built from technical data and prices only

Problem Statement

Batteries buy energy when it is cheap and sell it when it is expensive. What they can sell depends on what they stored before: their **state of charge (SOC)**. (They dont have a clear marginal cost).

Today, the independent system operator (**ISO**) clears each hour from the batteries' bids and leaves the SOC to the owners.

Baldick [1] proposes that owners submit their battery's parameters and that the ISO schedules the SOC: efficient, but it needs a market reform.

Research Gaps and Questions

Gap: how should batteries bid in today's market to reach the outcome of an SOC-based market, and with what information?

RQ1 Can today's bids give exactly Baldick's prices and schedules? What must the owner know?

RQ2 What is lost when the bids rely on imperfect price forecasts?

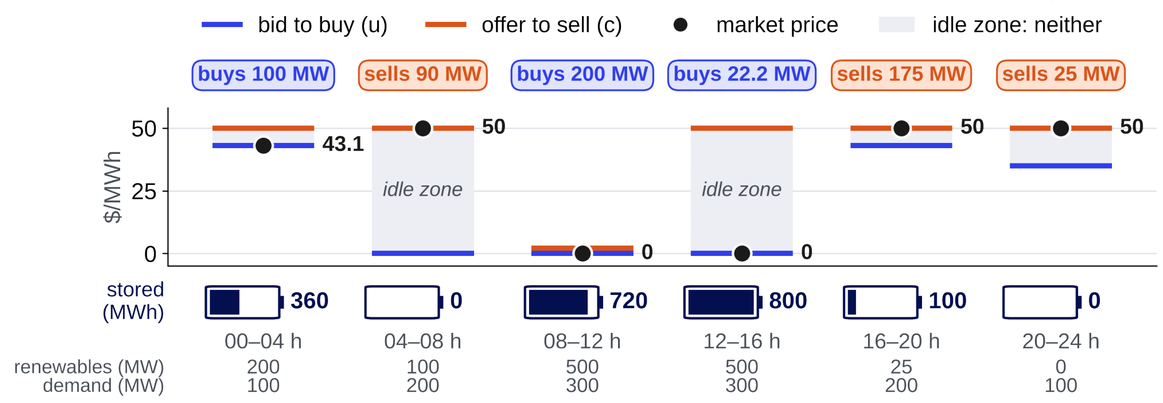
RQ3 Can the ISO build the same bids to detect market power?

Case Study: Baldick's Example

One bus, six 4-hour intervals; $\alpha = 0.9$, $\beta = \gamma = 1$, $\varepsilon = \zeta = 1$ \$/MWh, $\bar{s} = 800$ MWh, $w = 40$ \$/MWh.

hour t	1	2	3	4	5	6
price λ* ($/MWh)	43.1	50	0	0	50	50
charge (MW)	100	0	200	22.2	0	0
discharge (MW)	0	90	0	0	175	25
energy stored (MWh)	360	**0**	720	**800**	100	**0**
bid u ($/MWh)	**43.1**	**0**	**0**	**0**	**43.1**	**35**
offer c ($/MWh)	**50**	**50**	**2.11**	**50**	**50**	**50**

These bids clear today's market at exactly Baldick's prices and schedules.



Model: from State-of-Charge´s market to today's bids

Current design
Our results
Baldick's SOC design
Use with forecasts

WHAT THE SYMBOLS MEAN

- α charging efficiency
- β MWh drawn per MWh sold
- γ share of energy kept each hour
- ε, ζ wear cost: charge / discharge
- λ_t price at the battery's bus
- θ all battery parameters
- u_t, c_t bid to buy / offer to sell
- π_t value of stored energy
- $\bar{v}, \underline{v}$ extra when full / empty

1 · TWO DESIGNS

Current design (today)
e.g. the ERCOT (Texas) day-ahead market
- Each hour: a price to buy (its **bid** u_t) and a price to sell (its **offer** c_t).

Our question
Can today's interval bids give the same prices and the same schedules as Baldick's design?

…and what would the owner need to know to build those bids?

SOC-based design
proposed by R. Baldick [1]
- The owner submits its parameters θ and one price w: the value of energy left at the end of the day. ISO schedules the SOC with the rest of the system.

2 · COMPARING THEM

How we compare the designs

Both markets are cleared by the ISO as a **linear program (LP)**: cheapest schedule, and prices.

With partial derivatives we get its optimality (**KKT**) conditions; at the market prices, where no participant wants to change its schedule.

What does it do at price λ_t?

	u_t		c_t
Today	charge	idle	discharge
		set equal	
SOC	charge	idle	discharge
	$\alpha(\pi_t - \bar{v}_t) - \varepsilon$		$\zeta + \beta(\pi_t + \underline{v}_t)$

low price → high price

Equivalent bids

Bid and offer at Baldick's thresholds, every hour:

$u_t = \alpha(\pi_t - \bar{v}_t) - \varepsilon \qquad c_t = \zeta + \beta(\pi_t + \underline{v}_t)$

Same conditions: **same prices, same schedules**

Closed form:. (Only dependant on π)

$c_t = \zeta + \beta \max\{\pi_t, \pi_{t-1}/\gamma\} \mid u_t = \alpha \min\{\pi_t, \pi_{t-1}/\gamma\} - \varepsilon$

3 · OUR RESULTS

Can the owner compute these bids itself, from information it already has? Yes:

The battery's own problem gives π

Each owner can maximise its own profit at its local prices, given its parameters θ. The solution contains π, the same numbers as in SOC's market.

It needs only θ **and the local prices**: no data on others, the grid or the ISO, the bids depend only on the battery's parameters and its local prices: $(u, c) = \Phi(\theta, \lambda)$. $(u, c) = \Phi(\theta, \lambda)$.

4 · USE

in practice: replace the true prices λ^* by a forecast $\hat{\lambda}$

Inputs
parameters θ + price forecast $\hat{\lambda}$
the owner's or the ISO's

→ **Own optimisation**
one battery, small; gives π and v

→ **Bids**
$(u, c) = \Phi(\theta, \hat{\lambda})$
+ selection rule

→ **A · The owner bids in today's market**
Baldick's outcome without a market reform; exact with a perfect forecast, otherwise a welfare loss ΔSW (RQ2).

→ **B · Market monitoring by the ISO**
Reference bids from the ISO's own forecast, as CAISO's storage default energy bids, to flag market power (RQ3).

Main Takeaways

Interval bids reproduce Baldick's SOC-based outcome exactly.

They need only the battery's own parameters and its local prices. The gap between the designs is an **information gap**: price forecasts.

Future Work

RQ2 Build bids from price forecasts and measure the welfare lost (ΔSW) compared with Baldick's design.

RQ3 Compare submitted bids with ISO reference bids and with CAISO's storage default energy bids.

[1] R. Baldick, “A state-of-charge based formulation for storage participation in electricity markets: Technical reference,” arXiv:2606.14993, 2026. [2] Y. Chen, R. Baldick, “Battery storage formulation and impact on day ahead security constrained unit commitment,” IEEE Trans. Power Syst., 37(5), 2022. [3] CAISO Department of Market Monitoring, “Battery bid cost recovery and mitigation data,” 2024.

Acknowledgement

Co-authors: Jalal Kazempour, Lukas Alfred Stippel